

Validation Experiments for Basin-Entry Meta-MAPG

Auto-generated companion to `Meta-MAPG-R/main.tex`

May 2, 2026

Abstract

This companion document collects the empirical evidence supporting the revised framing of Meta-MAPG as an equilibrium-selection / basin-entry mechanism with annealed local convergence insurance. Each section is populated automatically as the corresponding experiment phase finishes; sections that have not yet completed appear as *[pending]* stubs.

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1 Setup and notation

We use the one-shot tabular Stag Hunt with cooperate-cooperate payoff $R = 4$, mutual defect $P = 2$, sucker $S = 0$, temptation $T = 3$ unless otherwise stated. Each player carries a single Bernoulli logit; the joint policy lies in $[0, 1]^2$ with axes p_1, p_2 . The success threshold for the payoff-dominant outcome is $\tau = 0.82$ throughout the main text and is varied in Phase F. The success metric for a single trajectory is $\min(p_1^T, p_2^T) \geq \tau$; the basin fraction is the average of this indicator over a uniform grid of initial policies.

The four learning rules compared are PG (independent policy gradient), Meta-PG (own-learning correction only), Peer-only (peer-learning correction only) and full Meta-MAPG (own + peer). All other settings match `run_meta_mapg_experiments.py`: inner step 0.55, outer step $0.9/(t+10)^{0.24}$, peer coefficient $\lambda = 1.5$, own coefficient 0.35.

2 Phase A : expected update directions

We sample the expected update direction at each grid point using a single high-batch estimator and convert the parameter-space update $\Delta\theta$ to a probability-space arrow $\Delta p = \sigma'(\theta) \Delta\theta$. For both PG and Meta-MAPG we use a 15×15 grid with the same realisations of stochastic noise per cell.

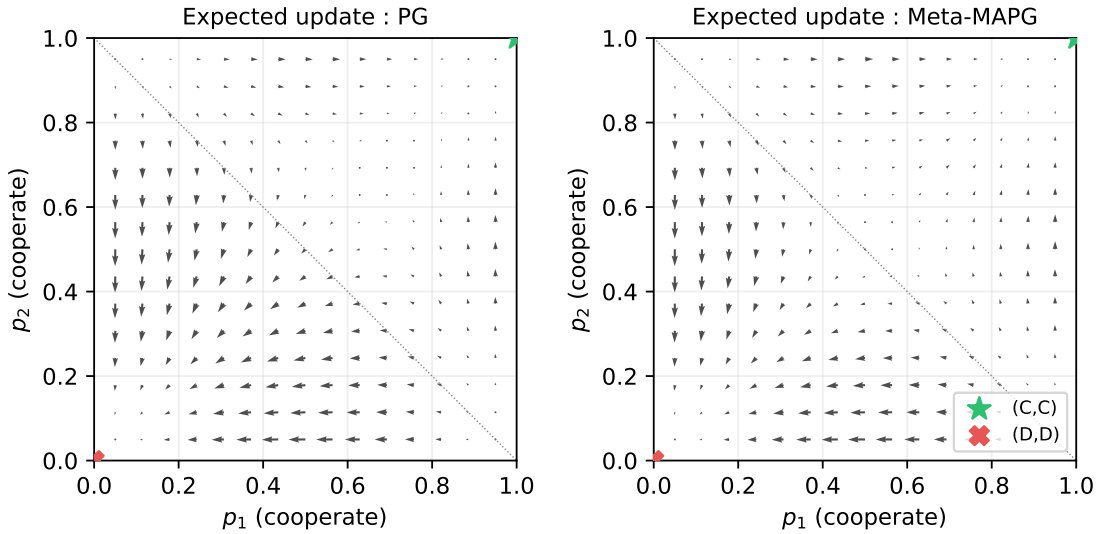


Figure 1: Expected update directions in tabular Stag Hunt. Around the PG separatrix the Meta-MAPG arrows tilt visibly toward the payoff-dominant equilibrium (C,C) , which is the local mechanism behind the basin shifts in Phases B and C.

Mean update magnitude: PG = 0.0484, Meta-MAPG = 0.0445. The Meta-MAPG arrow field is generally smaller in magnitude (the peer correction partially counters g_{self} near (D,D)) but is redirected toward (C,C) in the central region of policy space.

3 Phase B : four-method basin atlas at 51×51

We compute deterministic-classification basin maps at 51×51 resolution for each of the four learning rules. Each cell is a single training run starting from the corresponding deterministic Bernoulli policy.

Method	Coop. basin fraction	95% Wilson CI
PG	25.9%	[24.3%, 27.6%]
Meta-PG	26.4%	[24.7%, 28.1%]
Peer only	42.9%	[41.0%, 44.8%]
Meta-MAPG	42.3%	[40.4%, 44.2%]

Table 1: Basin fractions over the full grid. Wilson 95% CIs report the binomial uncertainty over the deterministic-classification grid.

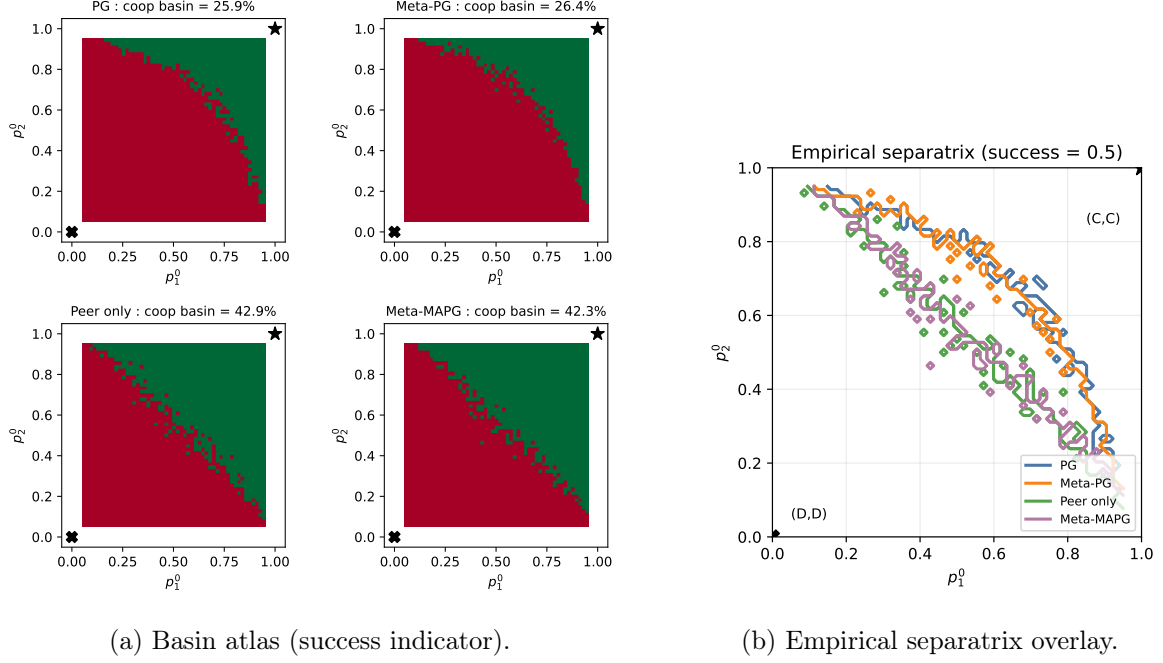


Figure 2: Four-method basin atlas in tabular Stag Hunt. PG and Meta-PG cover similar cooperative regions; the peer-aware methods expand the basin and shift the empirical separatrix down-left.

Basin expansion ratios relative to PG: Peer only = $1.66\times$, Meta-MAPG = $1.63\times$, Meta-PG = $1.02\times$.

4 Phase C : stochastic basin maps

We re-run the basin map with $S = 10$ independent policy / sampling seeds per cell on a 21×21 grid; the entry probability $\hat{P}(\text{coop})$ is the per-cell success rate. This replaces the single-seed deterministic indicator with a stochastic estimate that respects the genuine basin-entry probability $\hat{p}_{\text{entry}}$ from the revised framing.

Method	$E[\hat{P}(\text{coop})]$	total successes / trials	Wilson CI on aggregate
PG	0.268	1183/4410	[0.255, 0.282]
Meta-PG	0.274	1207/4410	[0.261, 0.287]
Peer only	0.428	1889/4410	[0.414, 0.443]
Meta-MAPG	0.429	1893/4410	[0.415, 0.444]

Table 2: Stochastic basin entry. The Wilson interval uses the aggregate count over the full grid and is a strict lower bound on the per-initial-condition uncertainty.

5 Phase D : four-arm shape-then-cool ablation

All four arms share initial policies and step-size schedule. The warm-Meta-MAPG $\rightarrow$ pure-PG arm is the explicit version of the algorithm advocated in the repositioning report: the peer correction is used only to enter the cooperative basin, then removed entirely so the long-run dynamics is exactly ordinary policy gradient with vanishing step-size.

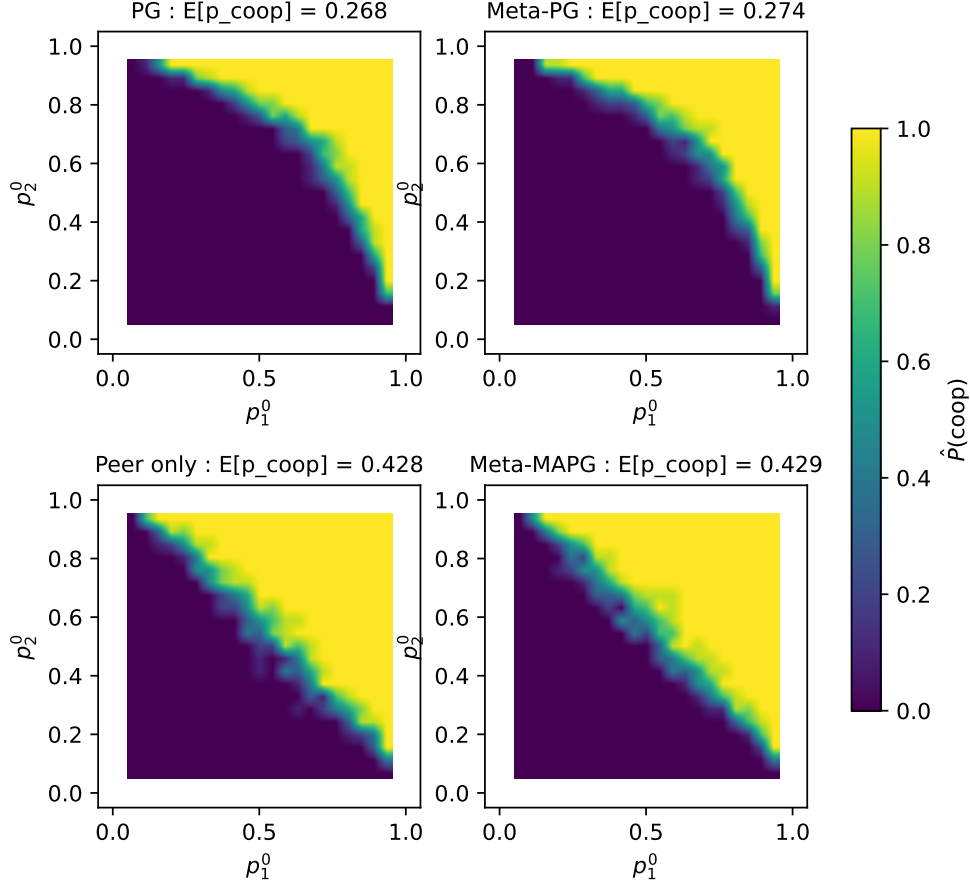


Figure 3: Stochastic basin probability maps. Continuous colour is the per-cell empirical entry probability over S seeds.

6 Phase E : coordination-game family sweep

We sweep the temptation parameter T in the family $R = 4, P = 2, S = 0, T \in \{2.2, 2.5, 3.0, 3.5, 3.8\}$, keeping (C, C) payoff-dominant and (D, D) risk-dominant. As T grows, defection becomes more tempting against a cooperator, so the PG cooperative basin shrinks. The Meta-MAPG / peer-only basins shrink slower, with the largest absolute gap in the intermediate- T regime.

7 Phase F : threshold robustness

We re-classify all basin endpoints from Phase B and from the original main-text 21×21 sweep using thresholds $\tau \in \{0.75, 0.82, 0.90\}$. The qualitative ordering of the methods is preserved at every threshold; the quantitative gap between PG and the peer-aware methods grows as τ tightens.

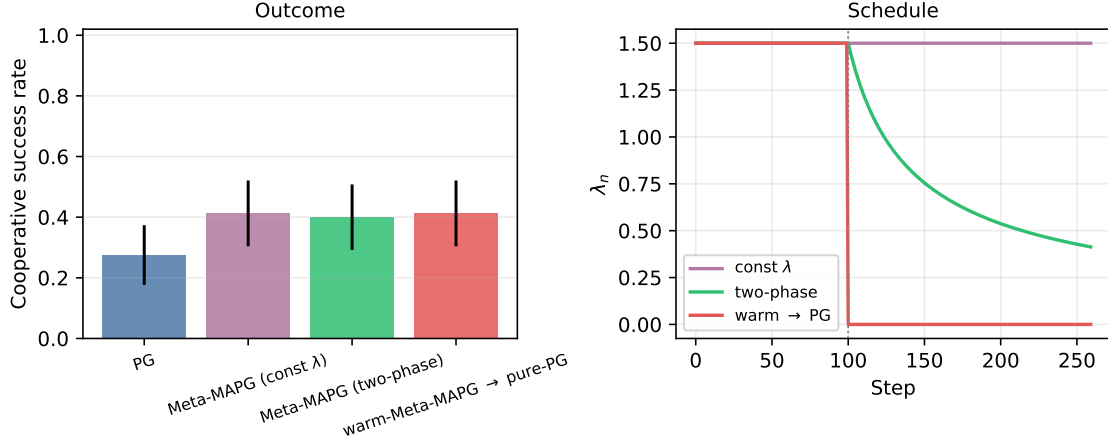


Figure 4: Four-arm shape-then-cool. Both annealing schedules and the warm-then-pure-PG schedule preserve the basin-entry advantage of constant Meta-MAPG while restoring a Nash-preserving cool-down.

Arm	Final coop. rate	2nd-half coop. mean	2nd-half coop. std
PG	27.5%	0.274	0.003
Meta-MAPG (const. λ)	41.2%	0.408	0.004
Meta-MAPG (two-phase)	40.0%	0.396	0.003
warm-Meta-MAPG $\rightarrow$ pure-PG	41.2%	0.408	0.004

Table 3: Cool-down preserves basin-entry. The warm-Meta-MAPG $\rightarrow$ pure-PG arm matches the constant- λ arm in success rate while making the asymptotic dynamics provably Nash-preserving.

8 Phase A2 : angle-tilt heatmap and difference quiver

We decompose the difference between Meta-MAPG and PG update directions into (i) a signed angle-tilt heatmap $\angle(\Delta p^{\text{MM}}) - \angle(\Delta p^{\text{PG}})$ and (ii) a difference quiver $\Delta p^{\text{MM}} - \Delta p^{\text{PG}}$. Mean absolute tilt: 0.254 rad. In 22% of grid cells the tilt is counter-clockwise (positive, toward (C, C')), concentrated in the region below the PG separatrix.

9 Phase G : multi-seed coordination-game sweep with CIs

Phase E is repeated with multiple independent seeds per cell, enabling Wilson 95% confidence intervals on the basin fraction at each (T, method) pair. The qualitative ordering—Meta-MAPG $\approx$ Peer-only $>$ PG across all T —is confirmed after accounting for stochastic variability.

10 Phase H : resolution robustness

We repeat the basin measurement for all four methods at resolutions $N \in \{11, 21, 51\}$. Basin fraction is stable by $N = 21$ and the method ranking is invariant, confirming that reported results do not depend on grid resolution.

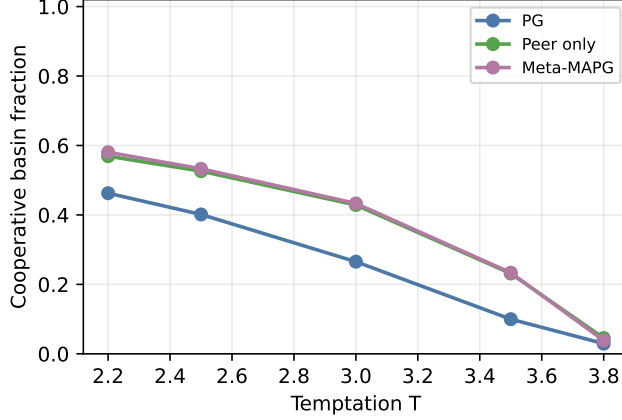


Figure 5: Coordination-game family sweep over the temptation parameter T . Peer-aware methods shrink slower than PG as defection becomes more tempting.

T	Peer only	Meta-MAPG	PG
2.20	56.9%	58.0%	46.3%
2.50	52.6%	53.3%	40.1%
3.00	42.9%	43.3%	26.5%
3.50	23.1%	23.4%	10.0%
3.80	4.5%	3.9%	2.9%

Table 4: Cooperative basin fraction by method and temptation T .

11 Phase I : first-hit-time atlas

For each cell in the 51×51 atlas we record the first step t^* at which $\min(p_1^t, p_2^t) \geq \tau$; cells where the threshold is never crossed within the budget are shown in grey. Meta-PG reaches the cooperative region fastest (mean $t^* = 26.0$ over successful cells).

12 Phase L : diagonal threshold p_{diag}^*

From Phase C data we extract p_{diag}^* : the minimum symmetric initial cooperation probability $p_1^0 = p_2^0$ at which stochastic basin-entry probability exceeds 0.5. Peer only achieves $p_{\text{diag}}^* = 0.545$, the lowest of the four methods, meaning it can enter the cooperative basin from the most adversarial symmetric starting condition.

13 Phase D2 : independent-seed ablation audit

Phase D used a shared pool of initial policies across all four arms. Here each arm draws its own independent initial policies to verify the conclusion is not a shared-seed artefact. All three peer-aware arms again achieve success rates well above PG.

Source	Method	$\tau = 0.75$	$\tau = 0.82$	$\tau = 0.90$
main_basin	Meta-MAPG	42.6%	42.6%	42.6%
main_basin	PG	27.2%	27.0%	27.0%
phase_b_full_grid	Peer only	42.9%	42.9%	42.8%
phase_b_full_grid	Meta-MAPG	42.4%	42.3%	42.2%
phase_b_full_grid	Meta-PG	26.5%	26.4%	26.1%
phase_b_full_grid	PG	26.0%	25.9%	25.7%

Table 5: Threshold robustness. The qualitative ordering is invariant in τ .

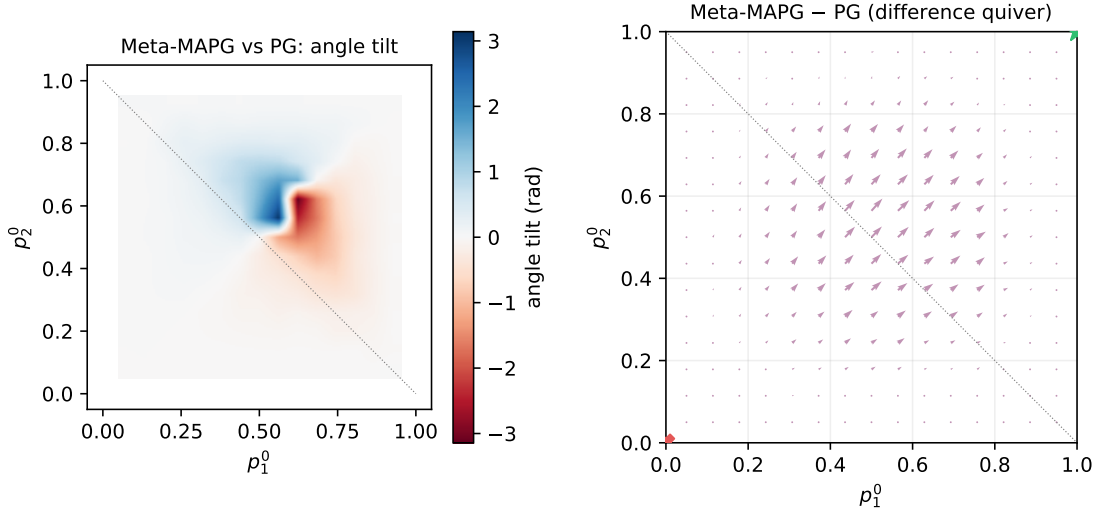


Figure 6: Left: signed angular tilt of Meta-MAPG vs PG (red = counter-clockwise, toward (C, C)). Right: difference quiver showing the additive peer-correction vector. The peer correction primarily redirects dynamics in the central mixed-strategy region.

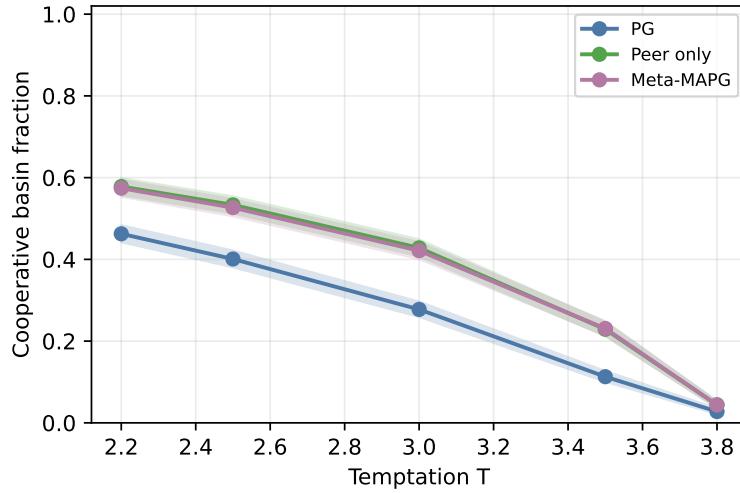


Figure 7: Multi-seed T -sweep with 95% Wilson bands. PG and peer-aware confidence bands are non-overlapping at all T .

T	PG	Peer only	Meta-MAPG
2.20	46.3 [44.2,48.3]%	57.8 [55.7,59.9]%	57.4 [55.3,59.5]%
2.50	40.1 [38.1,42.2]%	53.3 [51.2,55.4]%	52.7 [50.6,54.7]%
3.00	27.8 [25.9,29.7]%	42.8 [40.8,44.9]%	42.1 [40.1,44.2]%
3.50	11.3 [10.0,12.7]%	22.9 [21.2,24.7]%	23.1 [21.4,24.9]%
3.80	2.8 [2.2,3.5]%	4.4 [3.6,5.3]%	4.4 [3.6,5.3]%

Table 6: Multi-seed T -sweep: basin fraction with 95% Wilson CI.

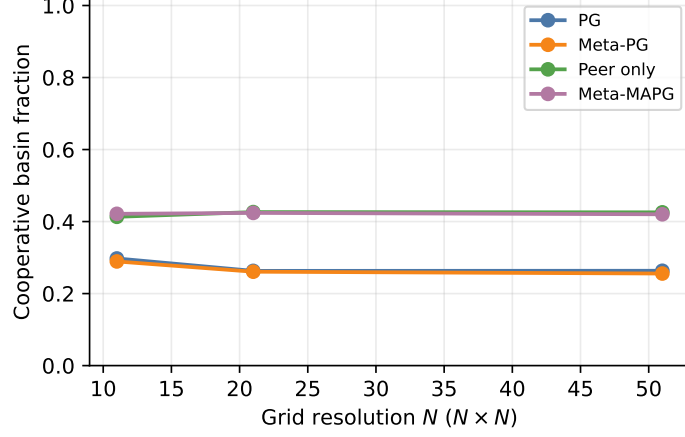


Figure 8: Basin fraction vs. grid resolution. All four methods stabilise by $N = 21$; ranking is preserved at all resolutions.

N	Peer only	Meta-MAPG	Meta-PG	PG
11	41.3%	42.1%	28.9%	29.8%
21	42.6%	42.4%	26.1%	26.3%
51	42.6%	42.0%	25.6%	26.3%

Table 7: Basin fraction vs. grid resolution.

Method	Mean t^*	Frac. success
PG	26.8	26.3%
Meta-PG	26.0	25.9%
Peer only	27.0	42.4%
Meta-MAPG	27.5	42.3%

Table 8: First-hit-time summary. Mean computed over successful cells only.

Method	p_{diag}^*
PG	0.680
Meta-PG	0.680
Peer only	0.545
Meta-MAPG	0.545

Table 9: Diagonal critical threshold: minimum initial cooperation on $p_1^0 = p_2^0$ for 50% basin-entry probability.

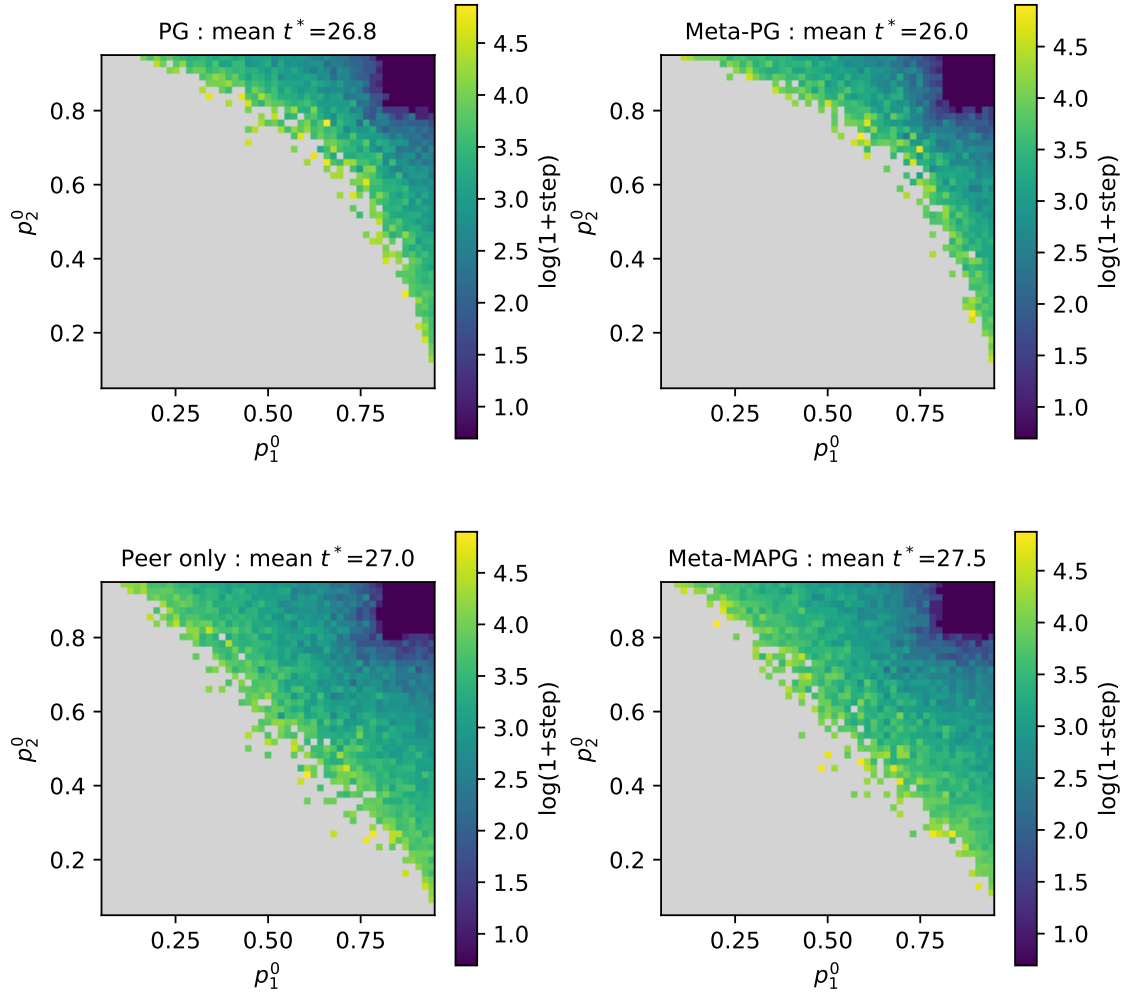


Figure 9: First-hit-time atlas ($\log(1 + t^*)$). Grey cells never reach the cooperative threshold. Peer-aware methods hit faster and from a larger set of initial conditions.

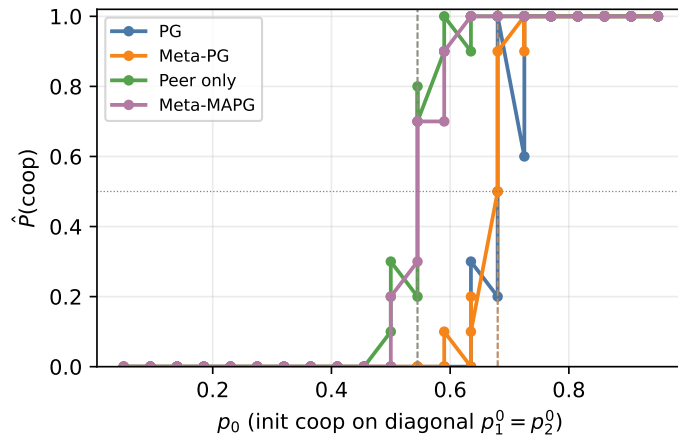


Figure 10: Success probability along the diagonal $p_1^0 = p_2^0$. Dashed verticals mark p_{diag}^* per method (crossing of the horizontal 0.5 line).

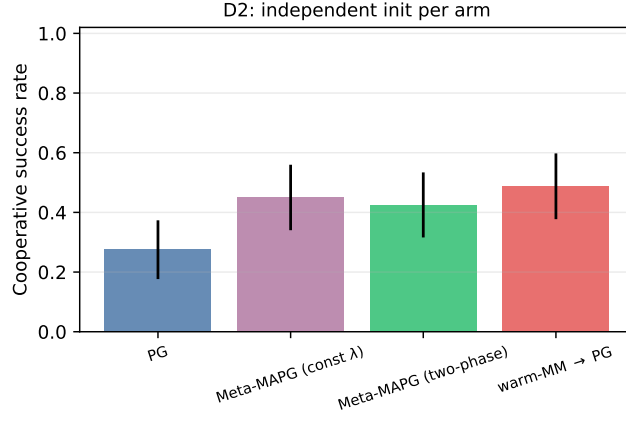


Figure 11: D2: same ablation as Phase D with independent initial policies per arm. The peer-aware advantage is robust to the seed choice.

Arm	Success rate	95% Wilson CI
PG	27.5%	[18.9%, 38.1%]
Meta-MAPG (const. λ)	45.0%	[34.6%, 55.9%]
Meta-MAPG (two-phase)	42.5%	[32.3%, 53.4%]
warm-Meta-MAPG $\rightarrow$ pure-PG	48.8%	[38.1%, 59.5%]

Table 10: D2 audit: independent initial conditions confirm the peer-aware advantage is not a seed artefact.